

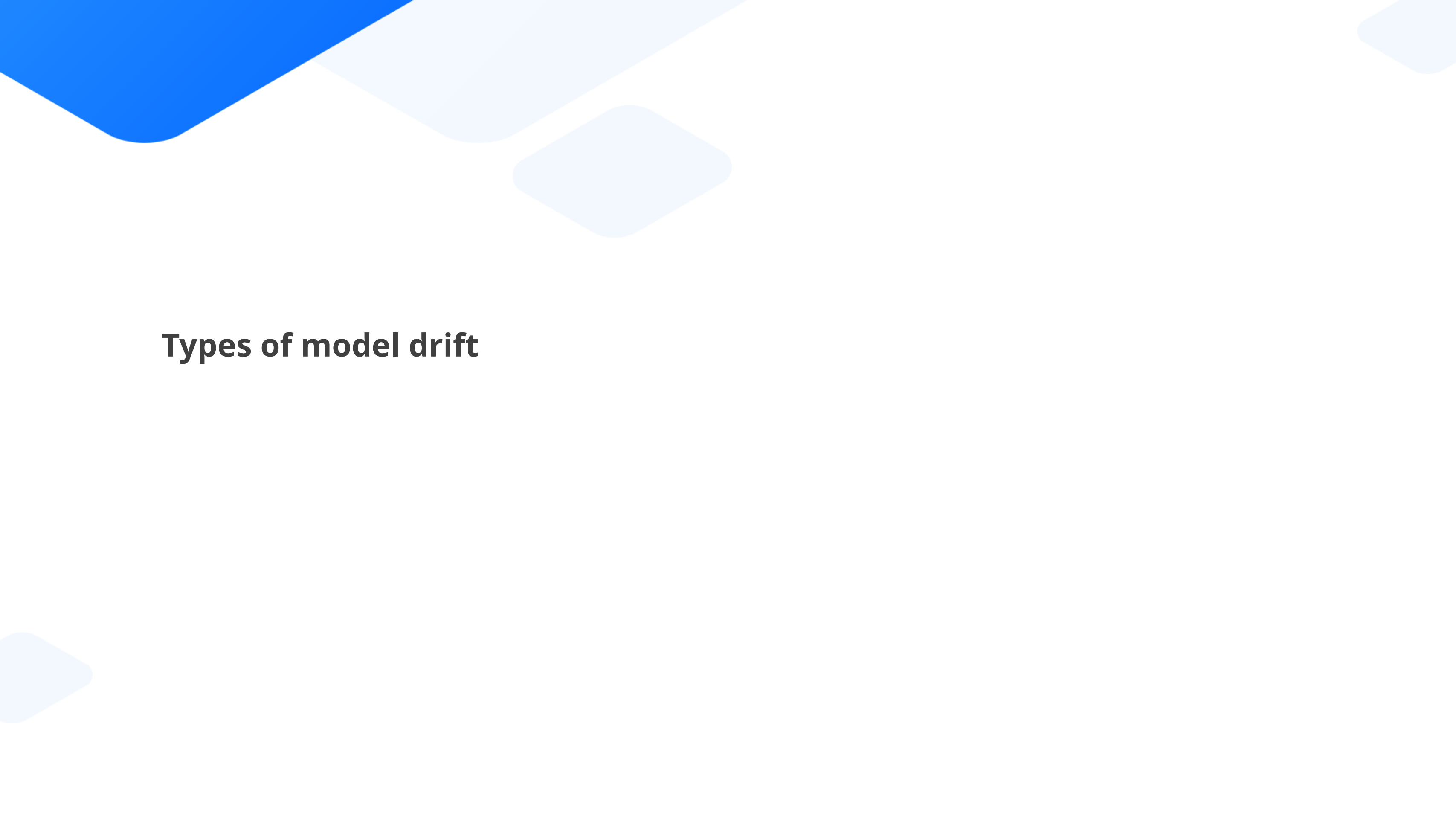
M07 - Data Drift and Concept Drift Detection



Topic

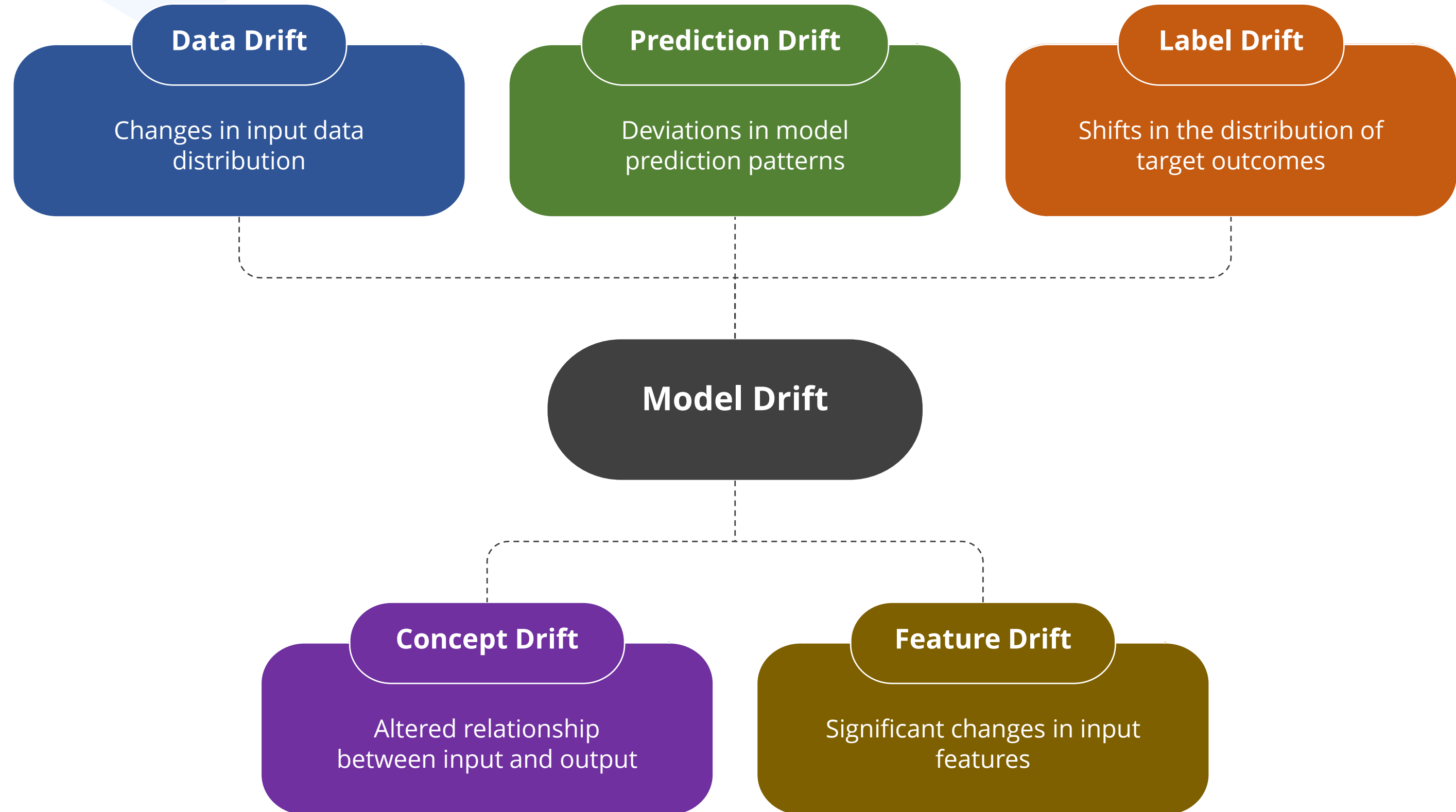
- Differentiate between data drift and concept drift in production AI systems.
- Identify and measure model drift using statistical techniques such as PSI, KS Test, and distribution analysis.
- Monitor model performance by configuring drift thresholds and automated alerts.
- Implement automated retraining workflows when significant drift is detected.
- Build reliable MLOps pipelines that continuously monitor, evaluate, and improve AI model performance in production.





Types of model drift

Types of model drift



1 DATA DRIFT



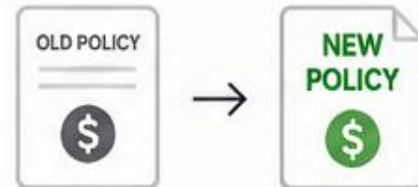
Input data changes
over time

USAGE_GB DISTRIBUTION



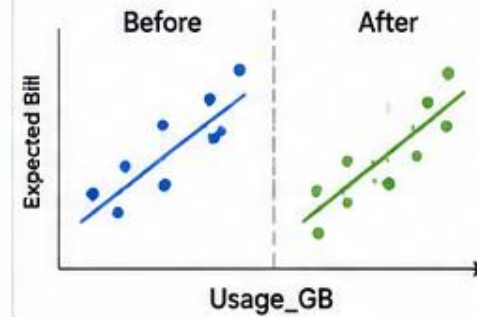
Impact
Model accuracy
drops

2 CONCEPT DRIFT



Relationship between inputs
and target changes

USAGE → EXPECTED BILL



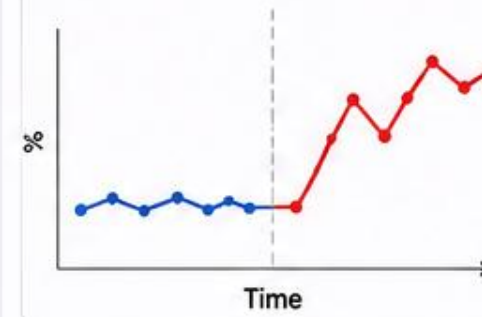
Impact
Business logic
has changed

3 PREDICTION DRIFT



Prediction pattern
changes over time

% ANOMALIES PREDICTED



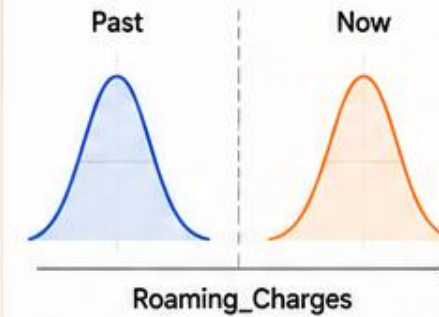
Impact
Predictions become
unreliable

4 FEATURE DRIFT



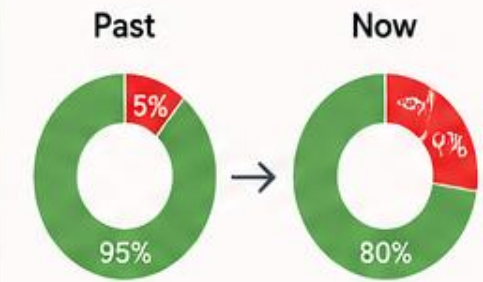
Input features change
significantly

ROAMING_CHARGES DISTRIBUTION



Impact
Key features shift,
affecting performance

5 LABEL DRIFT



Target (actual outcomes)
distribution changes

■ No Leakage (0) ■ Leakage (1)

Impact
Needs retraining
with new labels

FROM DRIFT DETECTION TO ACTION



1. MONITOR



2. DETECT



3. ALERT



4. TRIGGER RETRAINING



5. DEPLOY & IMPROVE

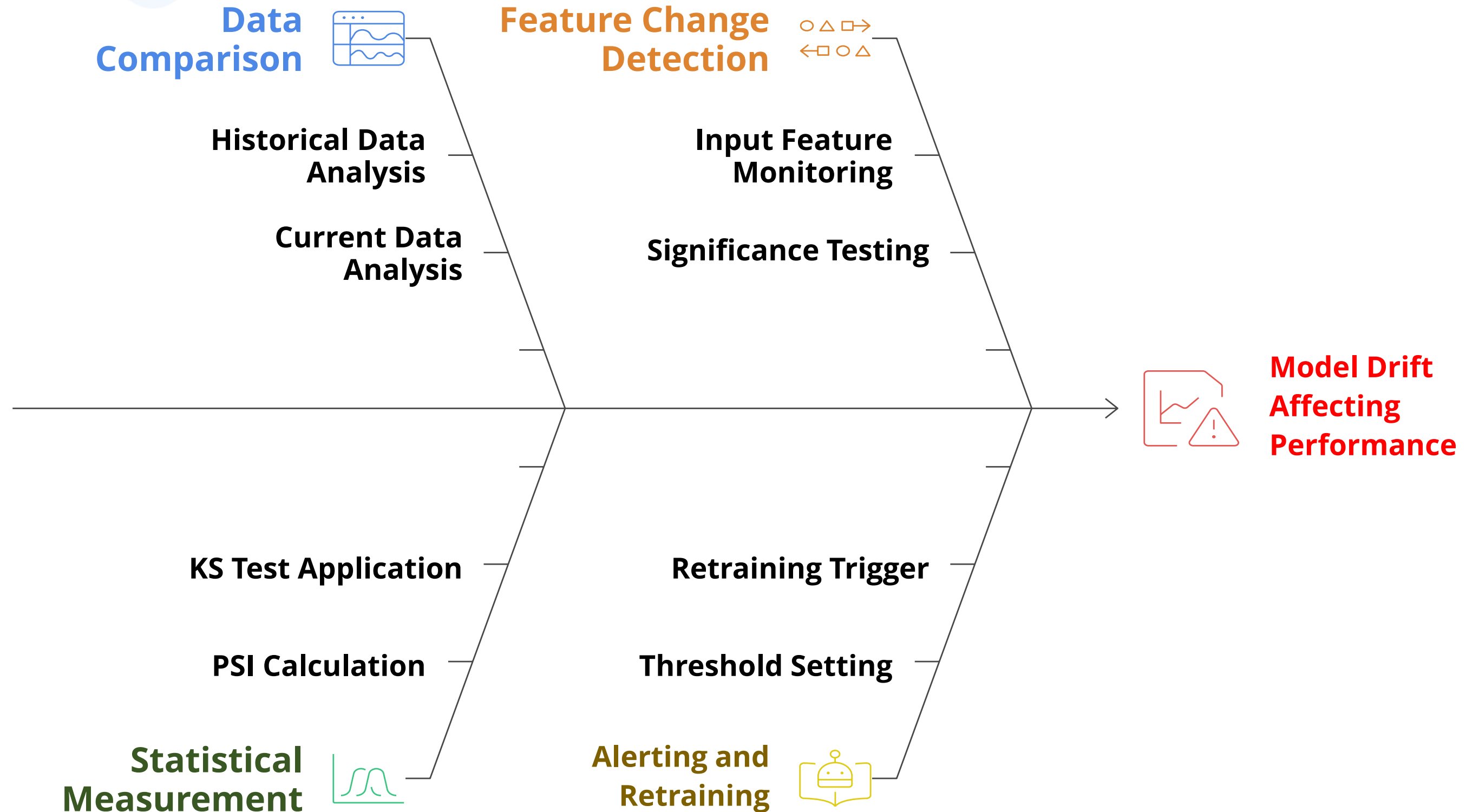


**ACCURATE MODELS
BETTER BILLING
HIGHER REVENUE**



Statistical drift detection techniques

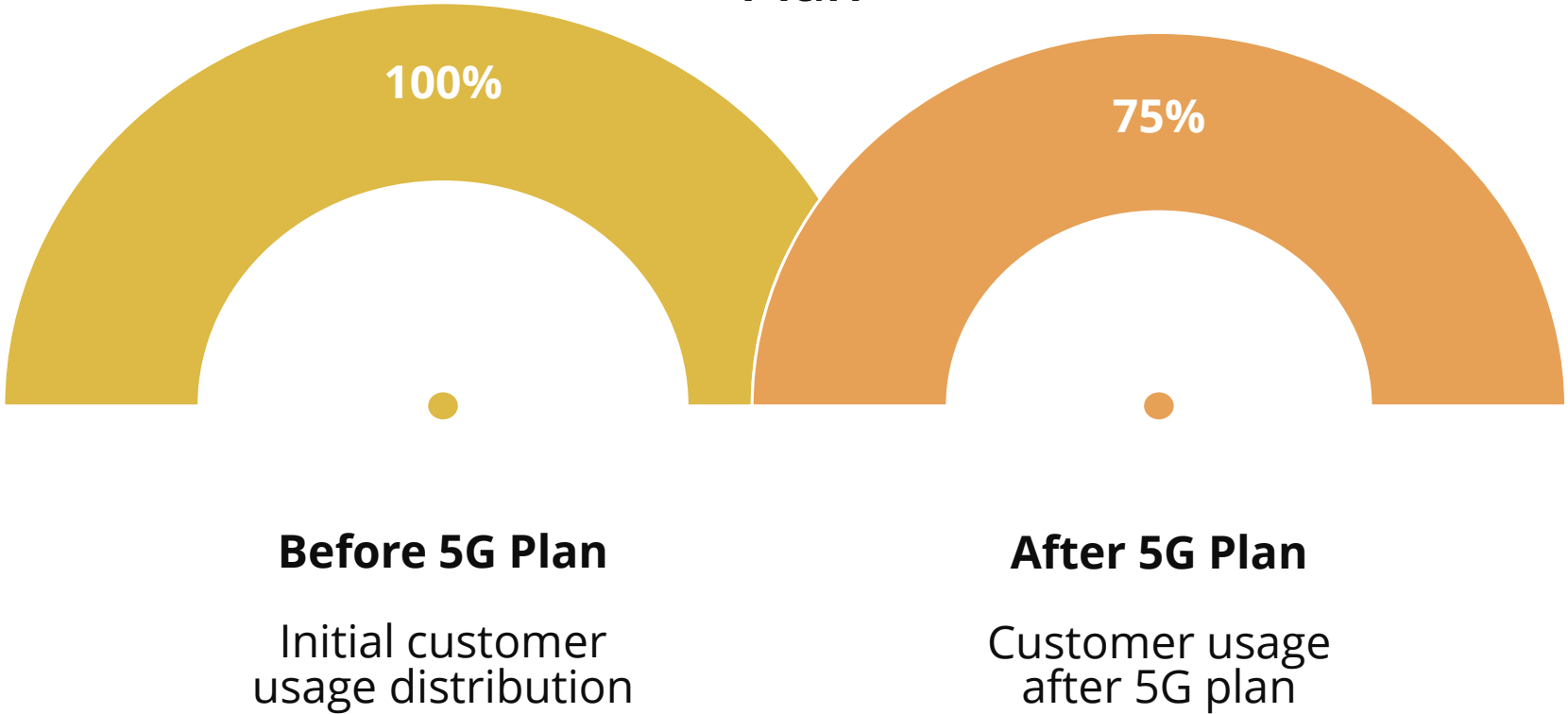
Statistical drift detection techniques



Statistical measurement techniques

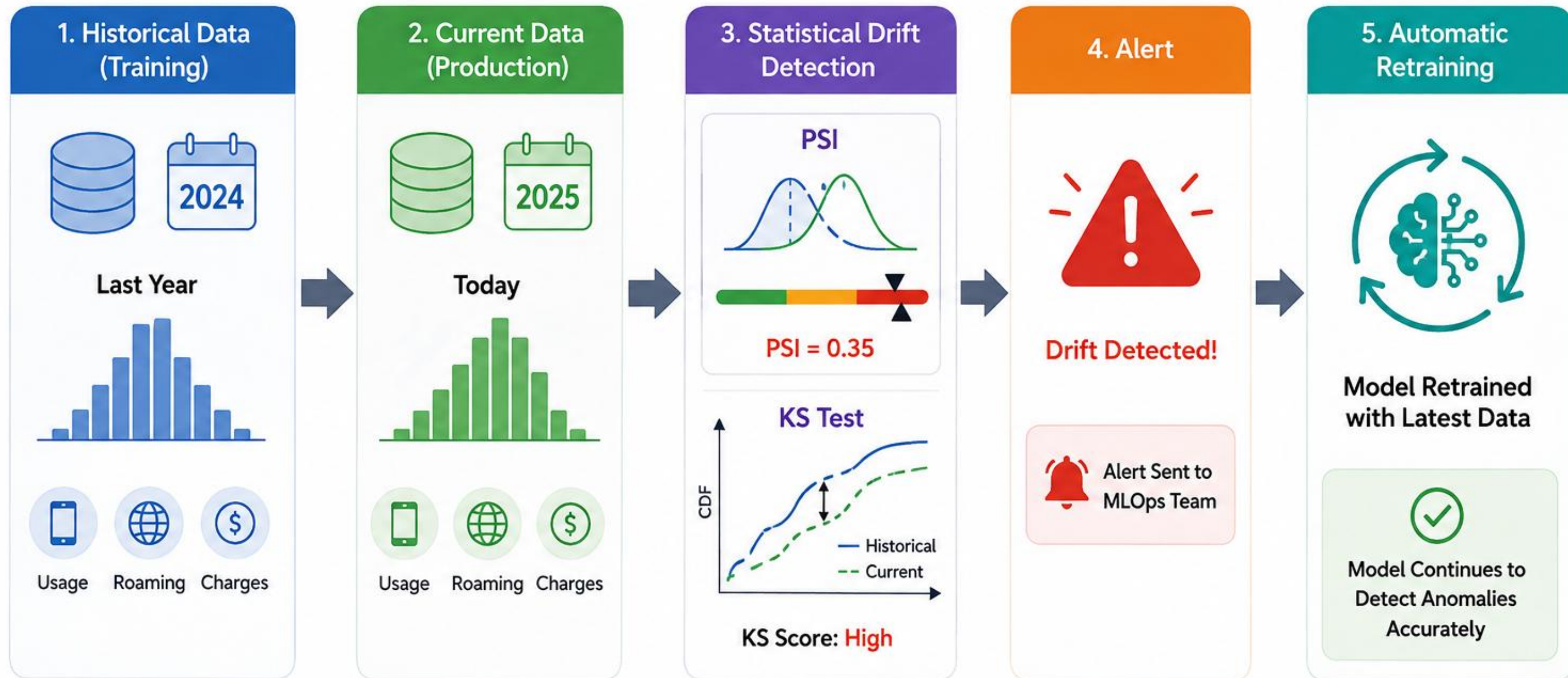
Technique	Full Form	What It Measures	Example
PSI	Population Stability Index	Measures how much input data has shifted over time	Average data usage changed from 8 GB → 15 GB
KS Test	Kolmogorov-Smirnov Test	Compares whether two data distributions are statistically different	Compare customer usage distribution before vs after a new 5G plan

Customer Usage Distribution Before vs After 5G Plan



How Statistical Drift Detection Works

Compare. Detect. Alert. Retrain.



Why Drift Happens?



New 5G Plans



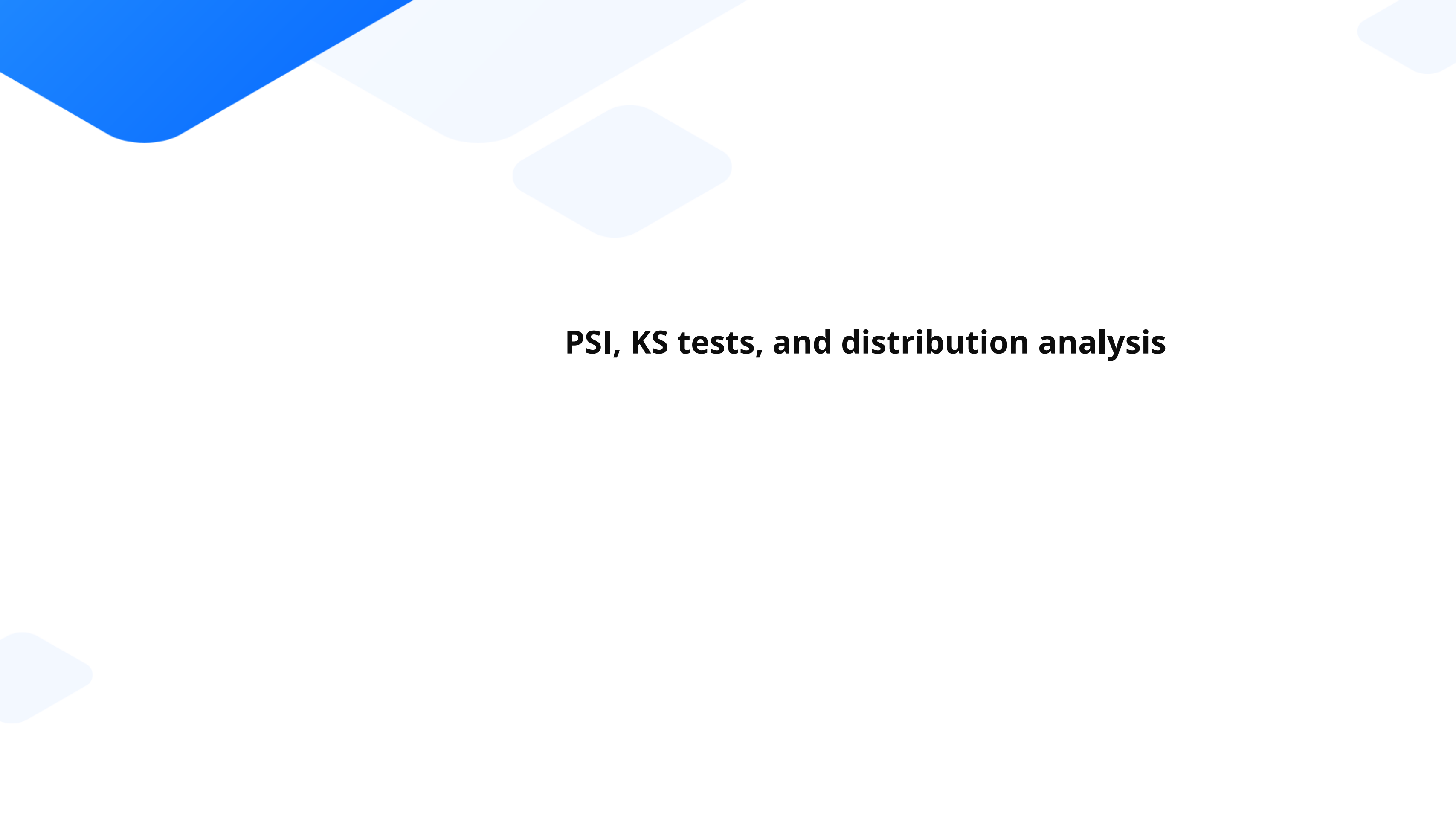
Roaming Offers



Seasonal Events

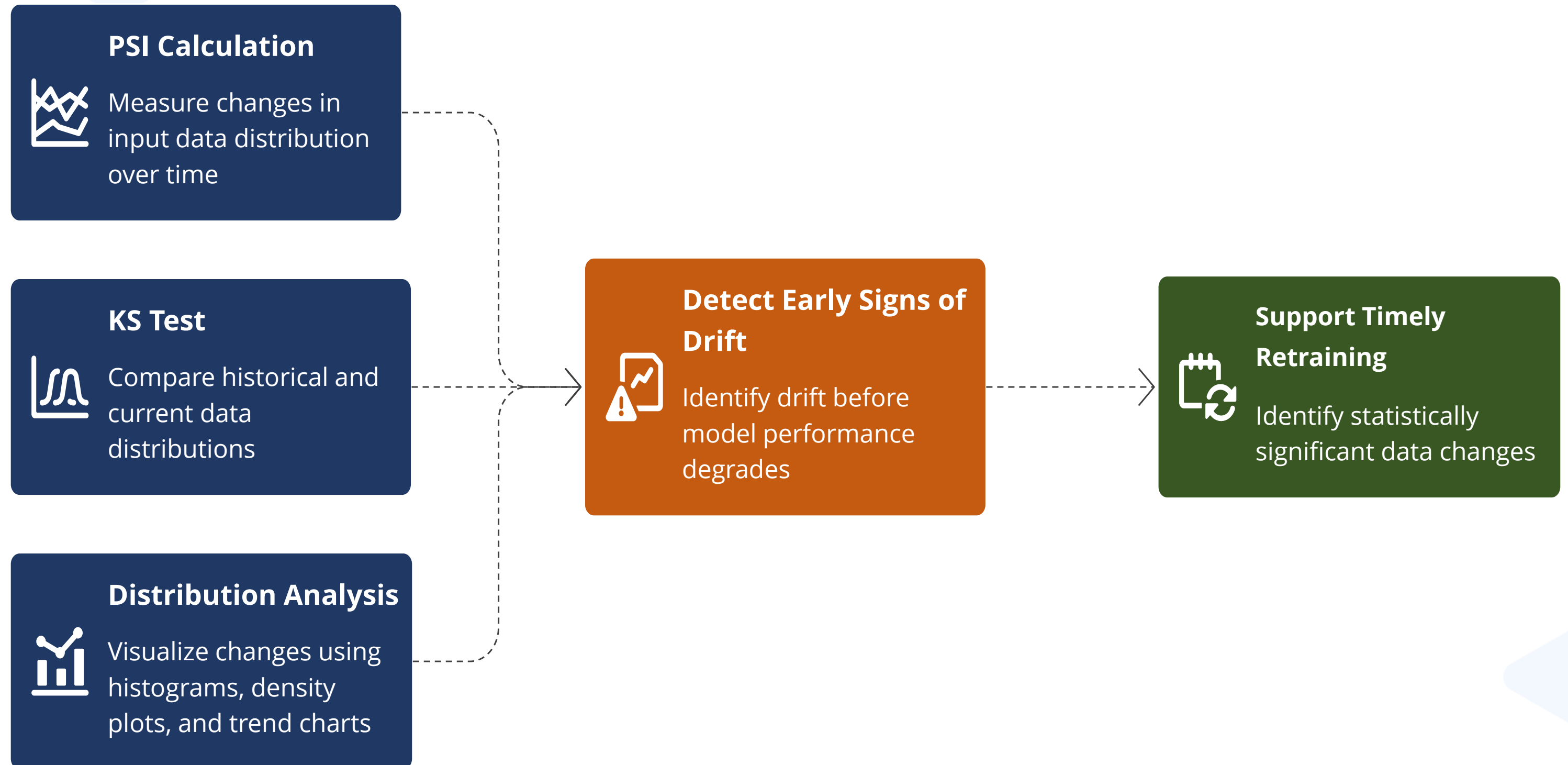


Accurate Billing.
Happy Customers.



PSI, KS tests, and distribution analysis

PSI, KS tests, and distribution analysis



Detect Data Drift with Key Techniques

Compare. Detect. Act.

1 PSI

Population Stability Index

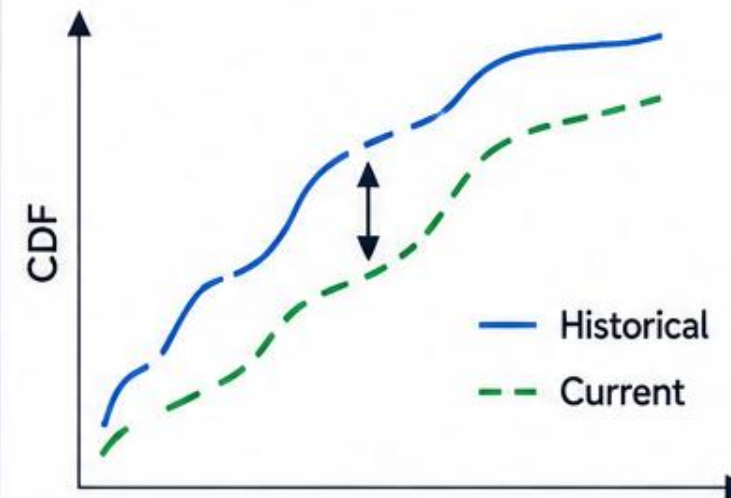


PSI = 0.35

Drift Detected

2 KS Test

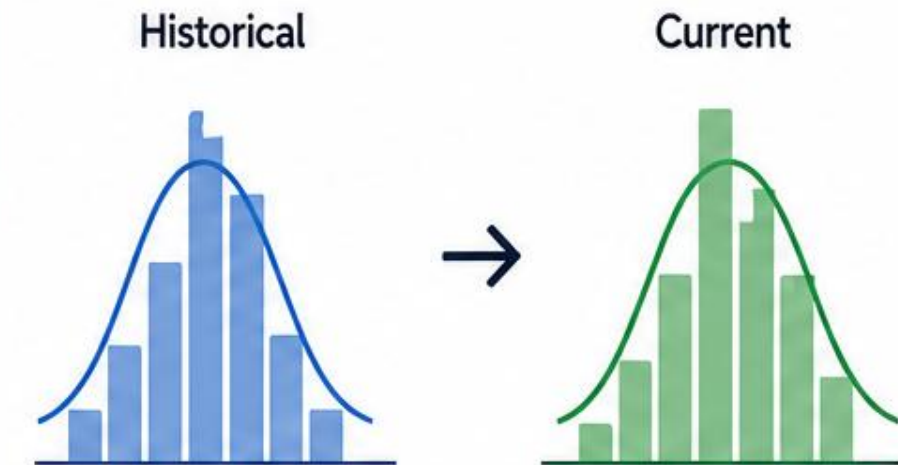
Kolmogorov-Smirnov Test



KS Score: High

Distributions are different

3 Distribution Analysis



Visualize changes
in data distribution



Data Changes



Detect Drift



Alert



Retrain Model

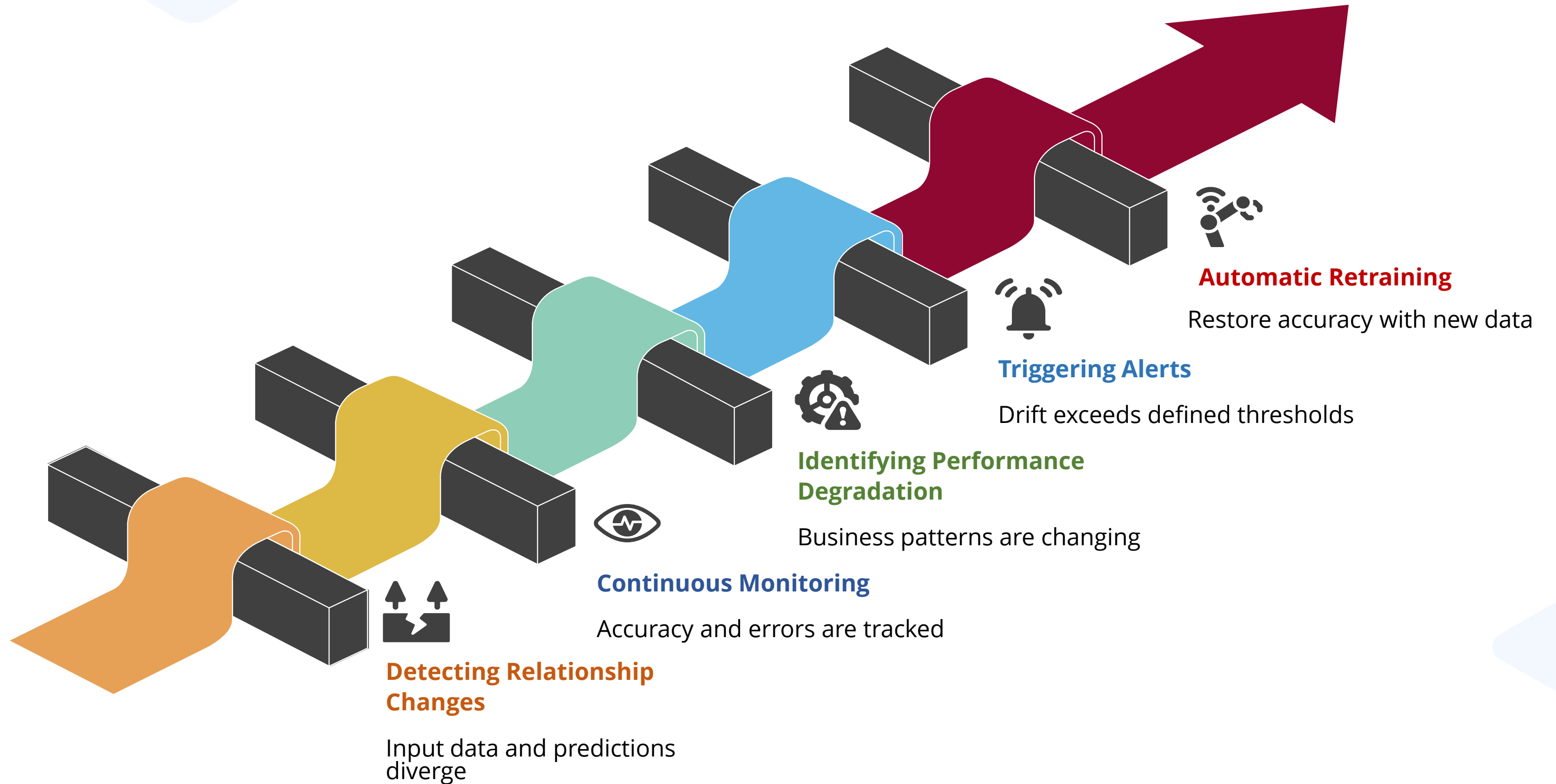


Accurate Billing
Better Experience

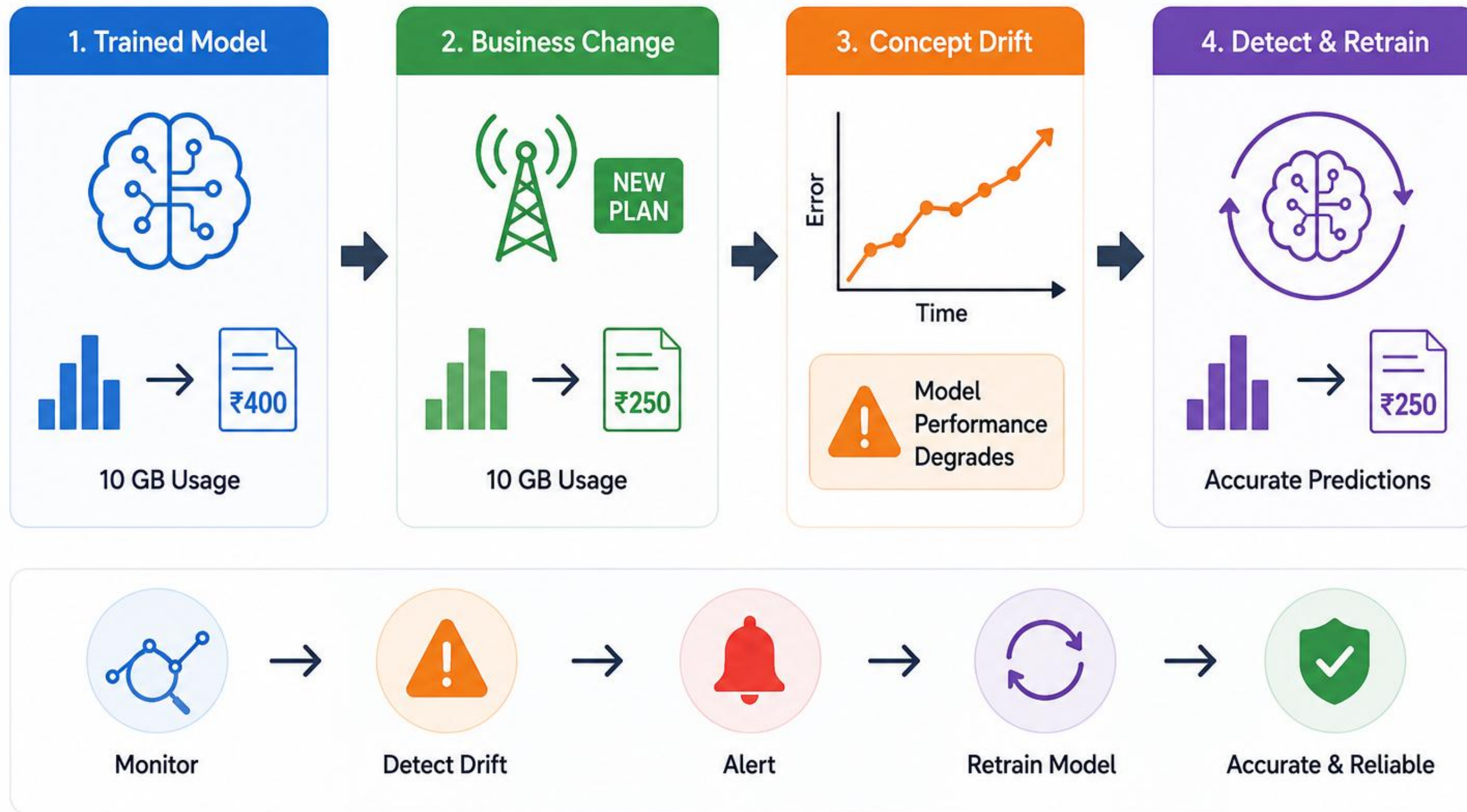


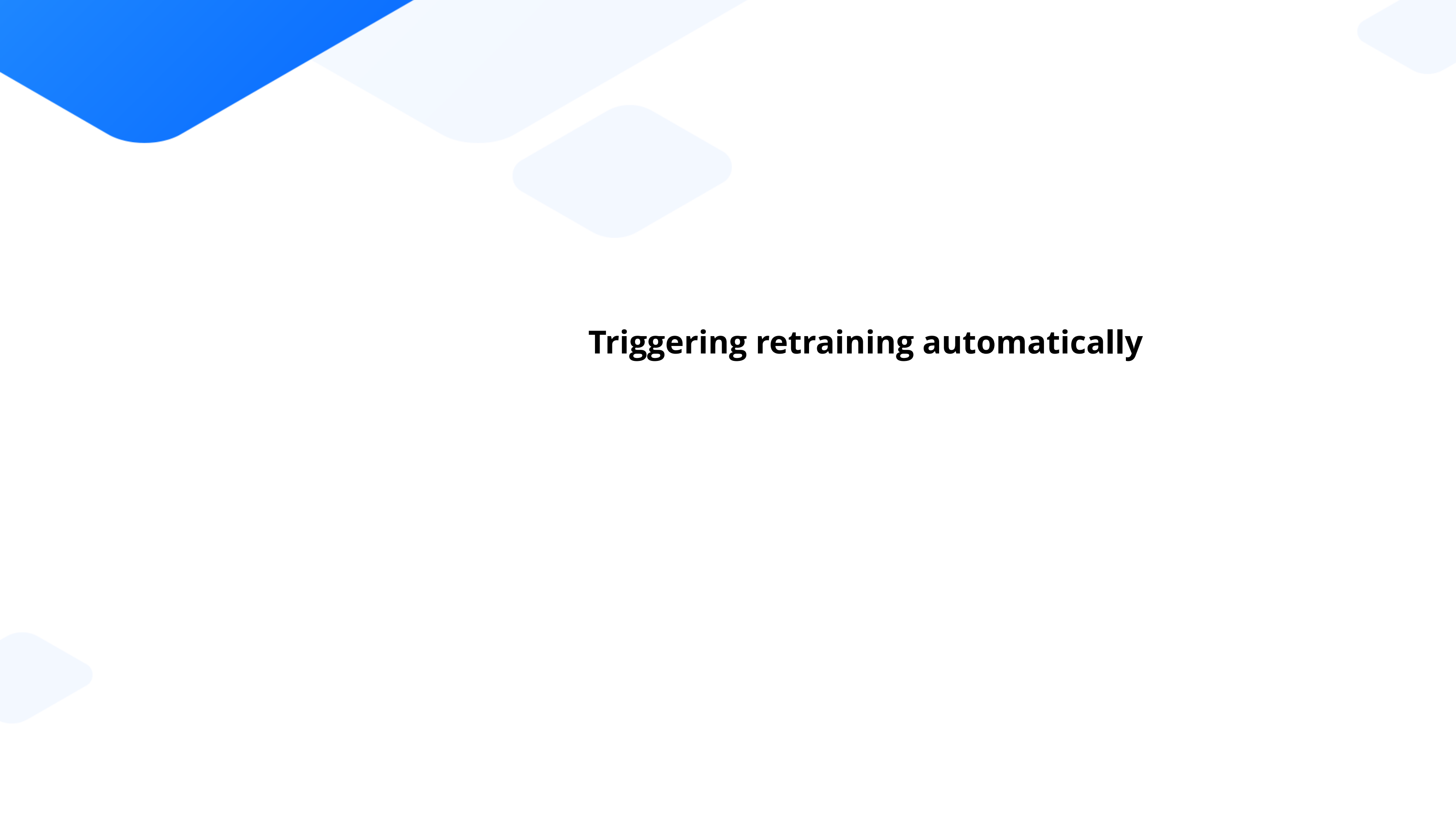
Concept drift algorithms

Concept drift algorithms



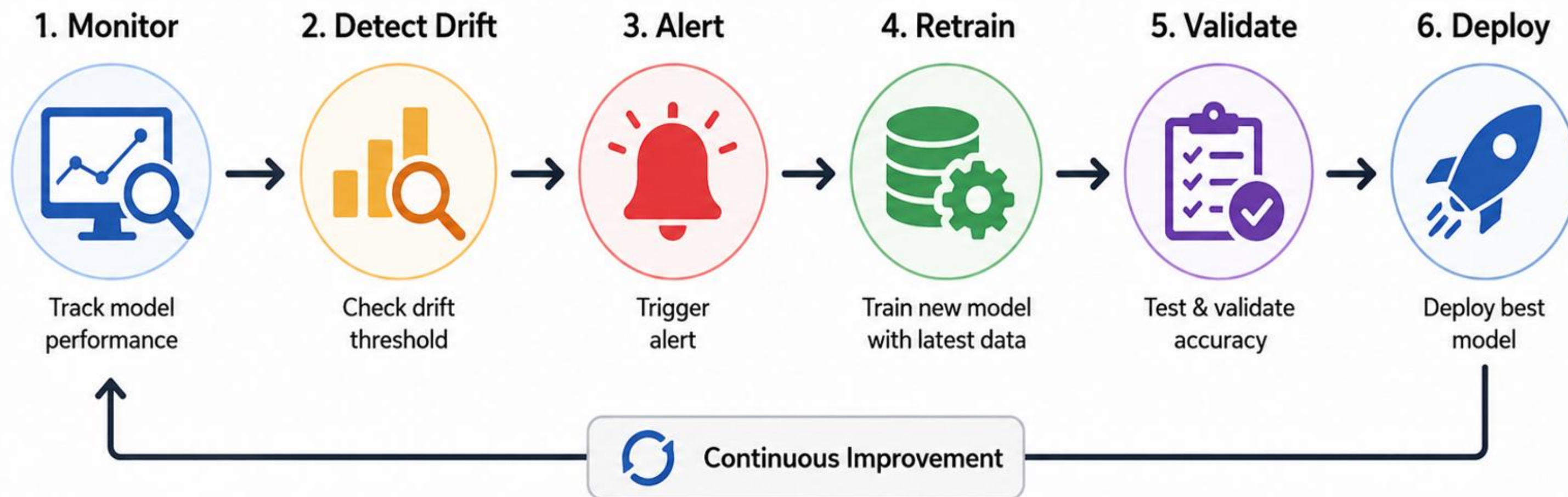
Concept Drift in Action



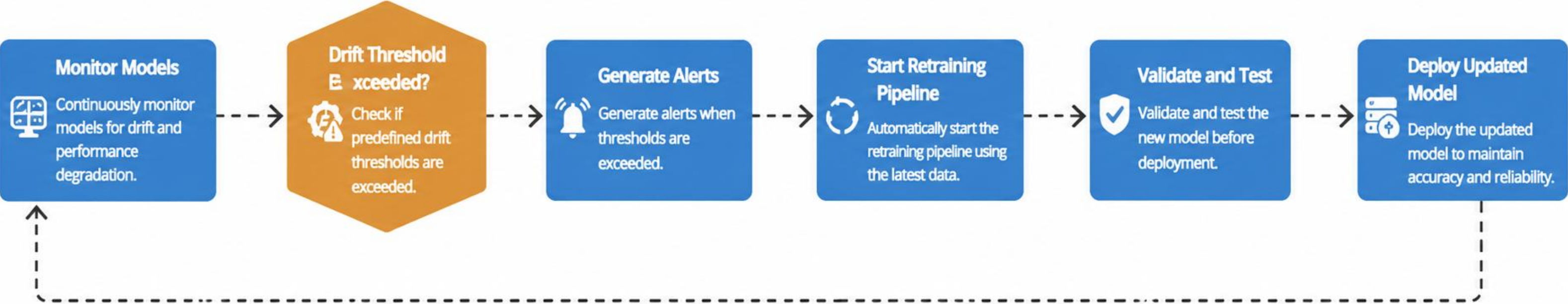


Triggering retraining automatically

Triggering retraining automatically

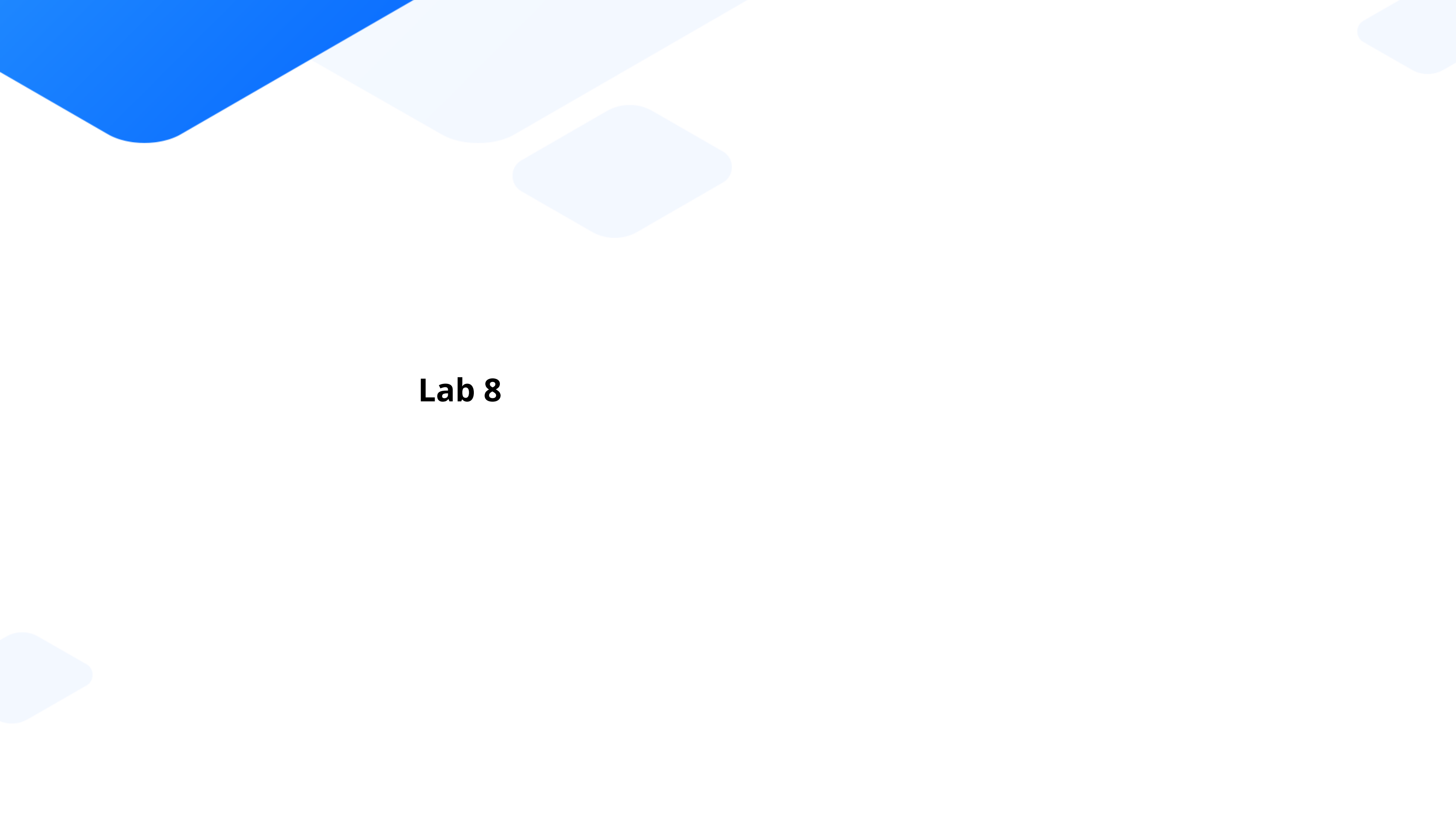


Triggering retraining automatically



Automated Model Retraining Workflow





Lab 8

QNA